

Muhammad Atif Zaheer

Address: 417 McAllister Building, State College, PA 16802, USA
Email: mvz5421@psu.edu
Website: mazaheer.github.io

Research Interests

Analytic number theory, particularly moments of L -functions, zero-free regions, and pair correlation of zeros of derivatives of completed L -functions.

Education

2024–: **Ph.D. in Mathematics**, Pennsylvania State University
Advisor: Amita Malik; expected completion: 2029
2022–2024: **M.S. in Mathematics**, Lahore University of Management Sciences
Advisor: Emanuel Carneiro
Thesis: *On Pair Correlation Conjecture and Bounds for the Averages of Montgomery's F -function*
2018–2022: **B.S. in Mathematics**, National University of Sciences and Technology

Papers and Preprints

2025 *Some Series Related to Extended Riemann Hypothesis for Dedekind Zeta Functions*, preprint, arXiv:2512.19761

Awards and Honours

2024: **Jack and Eleanor Pettit Scholarship in Science**, Eberly College of Science, Pennsylvania State University
2024: **National Management Foundation Gold Medal**, Lahore University of Management Sciences
2024: **Dean's Honour List**, Lahore University of Management Sciences
2022–2024: **Full Merit Scholarship**, Lahore University of Management Sciences
2022: **High Achiever Award**, National University of Sciences and Technology
2022: **First Place - Pakistan Calculus Contest**, IBA Sukkur University
2021: **Second Place - National Undergraduate Mathematics Contest**, Punjab University and National Mathematical Society of Pakistan
2019: **Honor Roll**, National University of Sciences and Technology

Teaching Experience

Instructor: **Pennsylvania State University**
Math 26: Plane Trigonometry, Fall 2025
Math 26: Plane Trigonometry, Spring 2026

Teaching Assistant: **Pennsylvania State University**
 Math 26: Plane Trigonometry, Summer 2025
 Math 220: Matrices, Summer 2025
 Math 441: Matrix Algebra, Spring 2025
 Math 441: Matrix Algebra, Fall 2024
Lahore University of Management Sciences
 Math 320: Algebra I, Spring 2024
 Math 230: Probability, Fall 2023
 Math 120: Linear Algebra, Summer 2023
 Math 205: Introduction to Analysis I, Spring 2023
 Math 101: Calculus I, Fall 2022

Talks

March 2026: *Littlewood's Proof of Prime Number Theorem and the Theory of Almost Periodic Functions*, Number Theory Student Seminar, Penn State University
 November 2025: *10^{12} Equivalent Statements of Riemann Hypothesis*, Number Theory Student Seminar, Penn State University
 May 2024: *On Pair Correlation Conjecture and Bounds for the Averages of Montgomery's F -function*, Thesis Defense, Lahore University of Management Sciences
 November 2023: *The Pair Correlation of the Zeros of the Riemann Zeta Function*, Graduate Student Seminar, Lahore University of Management Sciences
 June 2023: *Google Page Rank Algorithm*, Guest lecture for Math 120, Lahore University of Management Sciences

Conferences and Workshops Attended

September 2025: *MPS Conference on Universal Statistics in Number Theory*, Simons Foundation, New York
 March 2025: *Midwest Arithmetic Geometry and Number Theory Series (MAGNTS)*, Ohio State University
 March 2025: *The Legacy of John Tate, and Beyond*, Harvard University

Supervision

Spring 2026: Varnika Yadav & Kestian Apollos Orsetti, Directed reading project on prime number theorem

Professional Duties and Outreach

2025-2026: Organizer, Number Theory Student Seminar, Penn State University
 August 2025: Facilitator, Bootcamp session on abstract algebra's qualifying exam, Penn State University
 2023: Organizer, Salam Sessions for LUMS Students' Mathematical Society
 Summer 2023: Instructor, Math Circles – LUMS STEM Bootcamp, Government Boys High School (Sukamaidan) Skardu, Gilgit Baltistan

Miscellaneous

Mentioned in the preface of fourth editions of texts *Analysis I* and *Analysis II* by Terence Tao

Accepted into fully funded IAS/PCMI Graduate Summer School, 2024 (did not attend)

Accepted into fully funded Duke Number Theory School, 2026 (did not attend)